Network Scanning Report

Week 2 | Project Module 5 — Zenmap

Target: Local LAN subnet 192.168.100.0/24

Prepared on: Windows 10/11, Zenmap (Nmap 7.993)

Scope: Own home/LAN network — authorized self-scan

For educational purposes only — authorized lab exercise

1. Background & Objective

Zenmap is the official graphical front-end for Nmap, the industry-standard network scanning tool used by cybersecurity professionals and penetration testers alike. It wraps Nmap's command-line engine in a GUI, making host discovery, port scanning, and network topology mapping accessible to beginners while still exposing the full power of Nmap to advanced users.

This report documents a local area network discovery exercise performed with Zenmap against the author's own home LAN subnet. Unlike the earlier remote footprinting tasks performed against networkwalks.com, this scan touches the network directly using ICMP and ARP probes, which is why it was scoped strictly to an authorized, self-owned network rather than any third-party target.

Environment

- OS: Windows 10/11
- Tool: Zenmap (Nmap 7.993)
- Target subnet: 192.168.100.0/24 (home Wi-Fi LAN)
- Scan type: Ping scan (nmap -sn) — host discovery only, no port scanning

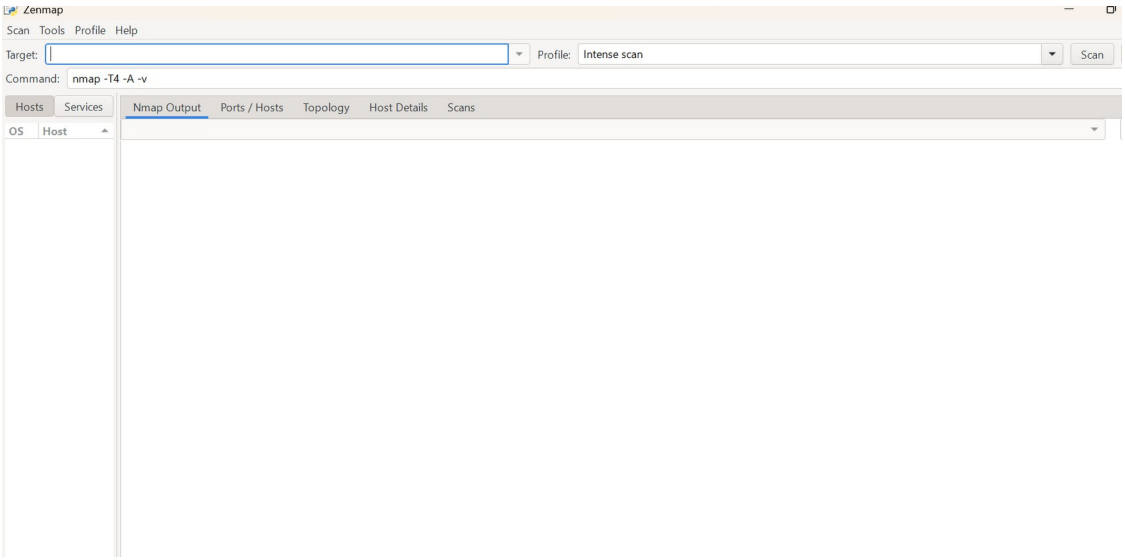
Task 1: Install Zenmap

Download and install Zenmap from the official Nmap website on a Windows PC.

Steps

- Downloaded the Windows self-installer from https://nmap.org/download.html
- Ran the installer, accepted the license agreement
- Kept all components selected (Nmap Core Files, Npcap, Zenmap GUI, Ndiff, Ncat, Nping)
- Completed the bundled Npcap driver installation
- Launched Zenmap from the desktop shortcut

Evidence



Zenmap launched successfully, confirming a working installation of Nmap 7.993 with the GUI front-end.

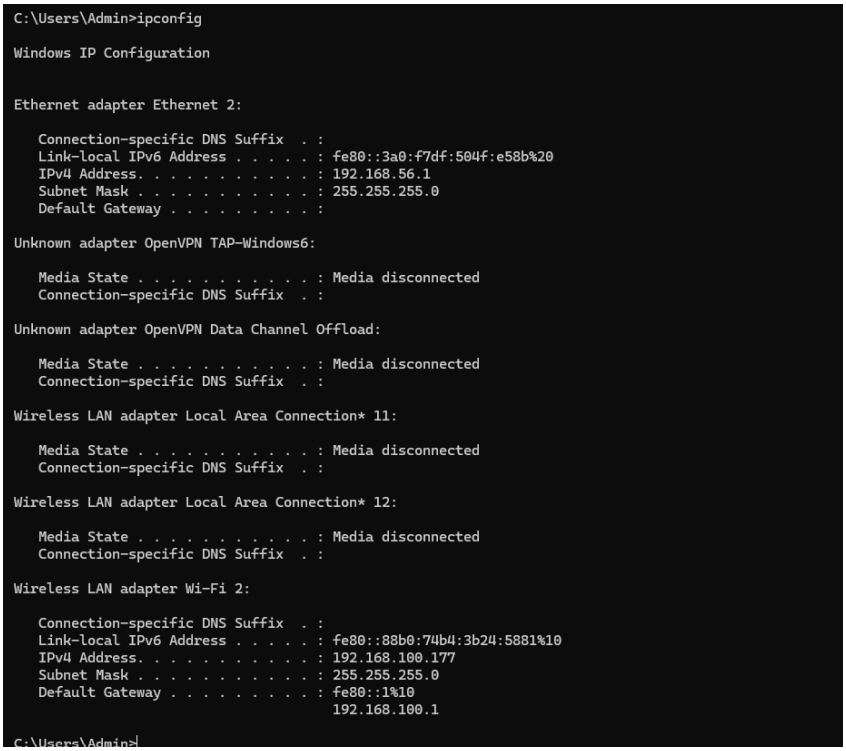
Task 2: Find Local IP Address & LAN Subnet

Find your local IP address and your LAN subnet.

Command

```
C:\> ipconfig
```

Evidence



Key Findings

Field	Value
Active adapter	Wireless LAN adapter Wi-Fi 2
IPv4 address	192.168.100.177
Subnet mask	255.255.255.0 (/24)
Default gateway	192.168.100.1
Derived subnet (CIDR)	192.168.100.0/24

Note: ipconfig also showed a second adapter, "Ethernet 2" (192.168.56.1), which is a virtual host-only adapter created by local virtualization software (e.g. VirtualBox) and has no default gateway. It was excluded since it does not represent the real LAN connection.

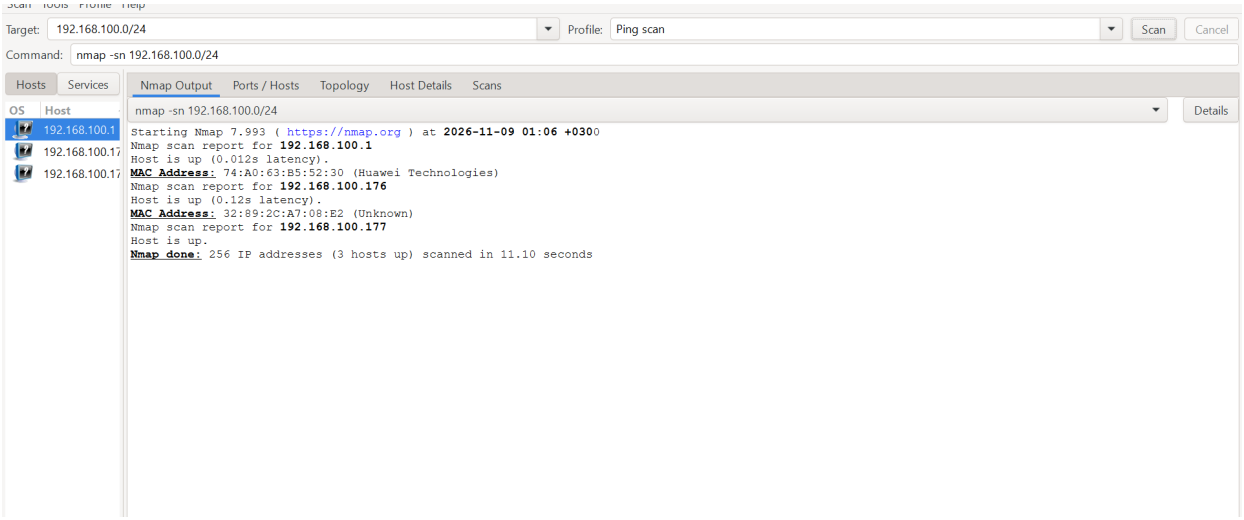
Task 3: Find Live Hosts in the Subnet

Open Zenmap, input the local LAN subnet, and select Ping Scan to find the list of live hosts in the subnet.

Command

```
nmap -sn 192.168.100.0/24
```

Evidence



The ping scan completed in 11.10 seconds, checking all 256 possible addresses in the /24 range.

Tasks 4–6: Scan Results

Task 4 — How many hosts are live?

Answer: 3 hosts are live (including the scanning PC itself), out of 256 addresses scanned.

Task 5 — IP addresses of the live hosts

IP Address	Likely Role
192.168.100.1	Router / default gateway
192.168.100.176	Unidentified device on the LAN
192.168.100.177	Scanning PC (this machine)

Task 6 — MAC addresses of the live hosts

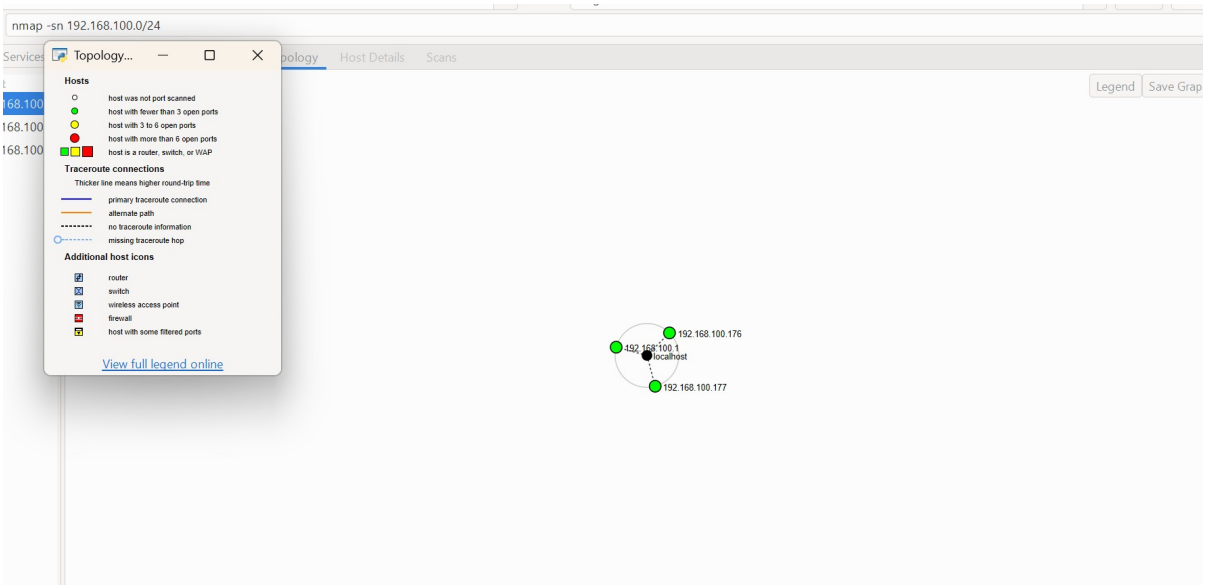
IP Address	MAC Address	Vendor
192.168.100.1	74:A0:63:B5:52:30	Huawei Technologies
192.168.100.176	32:89:2C:A7:08:E2	Unknown
192.168.100.177	(own PC — see ipconfig /all)	N/A

Note: Zenmap/Nmap does not report a MAC address for the scanning machine itself, since the host does not ARP its own interface the way it does for other LAN devices. The local MAC address must instead be read from ipconfig /all under "Physical Address."

Task 7: Topology Diagram

Display and save the output topology in PDF format.

Evidence



The topology view shows the three live hosts arranged around "localhost," each marked green (fewer than 3 open ports detected — expected, since a ping scan does not perform port scanning). The diagram was exported and saved as Zenmap_Topology.pdf to the Desktop.

Summary of Findings

- The home LAN subnet is 192.168.100.0/24, served by a router at 192.168.100.1 (Huawei hardware).
- A ping scan identified 3 live hosts in 11.10 seconds: the router, one unidentified device, and the scanning PC.
- MAC address vendor lookups partially identify device manufacturers even without port scanning, illustrating how much can be learned from host discovery alone.
- The topology view and legend provide a quick visual map of the network, useful as a starting point before deeper, authorized scanning (port and service scans) in later modules.

Why This Matters

Host discovery is typically the first active step a penetration tester or attacker takes once inside a network, or when targeting a specific subnet they have access to. Unlike the purely passive footprinting performed in W2-PM1 and W2-PM2 (which never touched the target directly), a ping scan does send packets to every address in the range, making it detectable by network monitoring tools or intrusion detection systems if run against a network the operator does not control.

This is why the scan in this report was deliberately scoped to the author's own home network rather than any third-party or shared network, consistent with the course's liability disclaimer: authorized use only, on devices or networks the person owns or has explicit permission to test.

Conclusion

This exercise demonstrated the full Zenmap workflow: installation, identifying the local network, discovering live hosts via a ping scan, extracting host and MAC address data from the results, and visualizing the network as a topology diagram. These are foundational skills for any later, more invasive scanning (port scans, service/version detection, OS fingerprinting), which should only ever be performed against networks and systems the tester is explicitly authorized to assess.